NetGist: Learning to Generate Task-based Network Summaries

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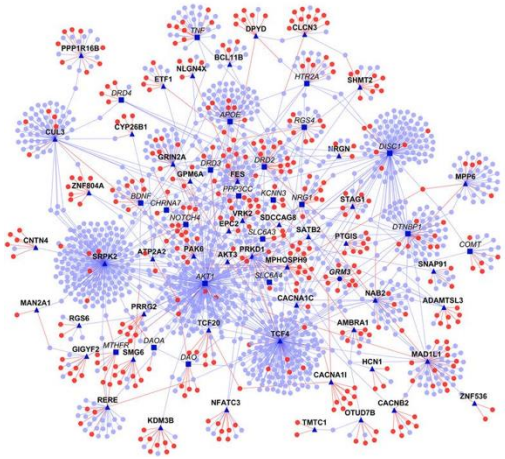
Virginia Tech

ICDM, Singapore, November, 2018

Real-world Networks

Facebook Network [2012]

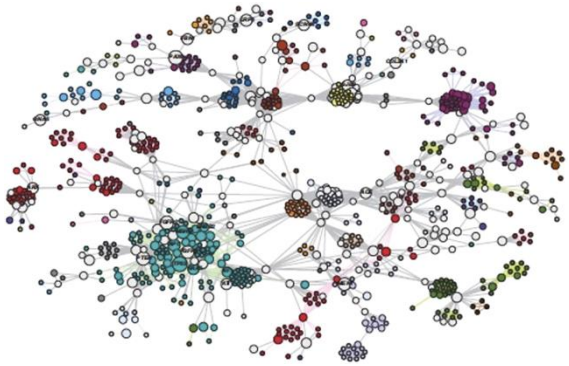




Protein interaction Network [Wikipedia 2016]

The Internet





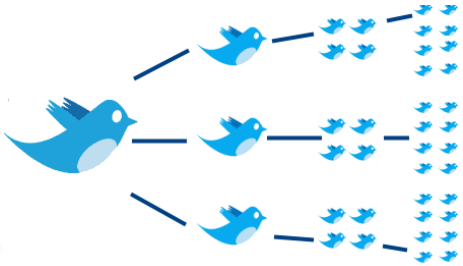
Human Disease Network [Kenneth 2011]

Graph Mining Tasks

- Viral Marketing
- Community Detection
- Information Diffusion
- Event Detection
- Anomaly Detection
- Immunization
- Pattern Detection

…





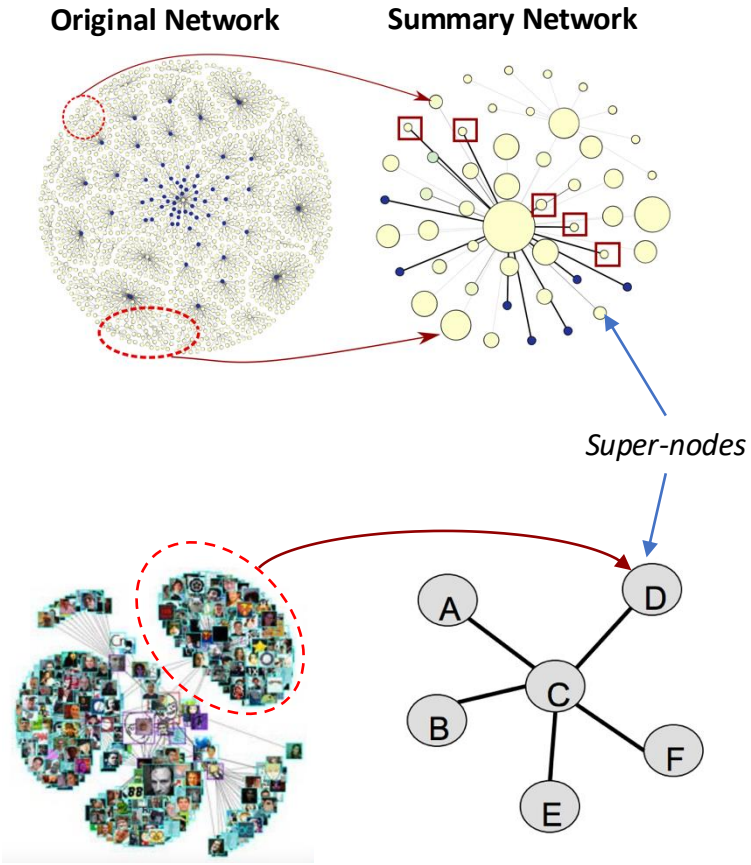




Problem: How to handle large scale networks?

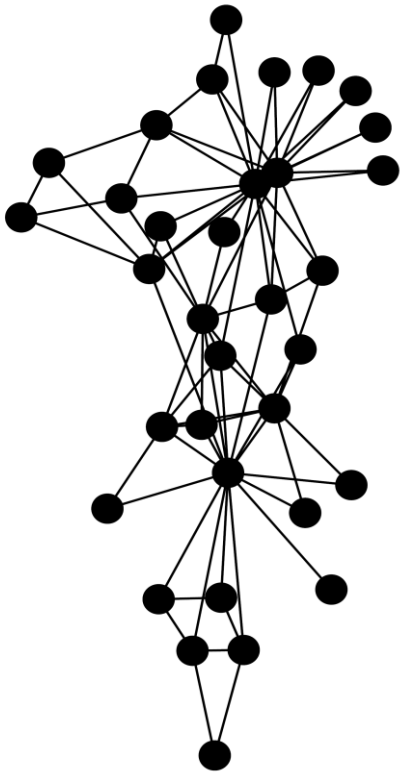
Graph Summarization

- A smaller graph that represents the original one.
- Useful for:
  - Visualization/Sense-Making [Koutra+, 2015]
  - Speeding up algorithms [Purohit+, 2014]
  - Anomaly detection [Amiri+, 2018]
  - …

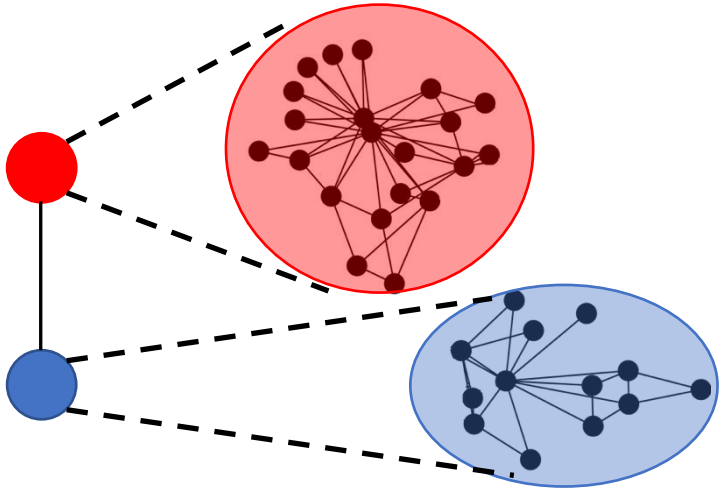


Summary is Task Dependent

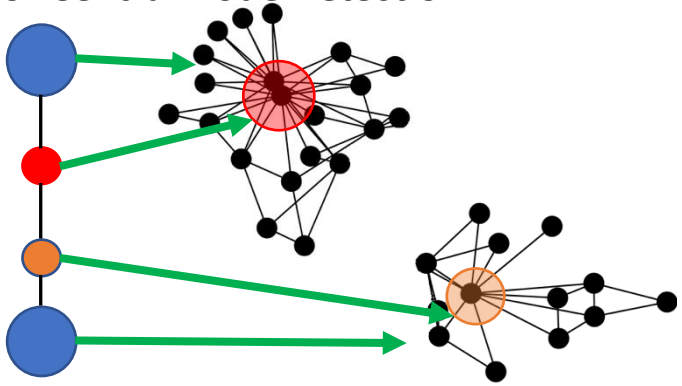
Karate Network



Summary for Community Detection



Summary for Central Node Detection



Prior Works on Summarization

- Prior graph summarization techniques are task specific

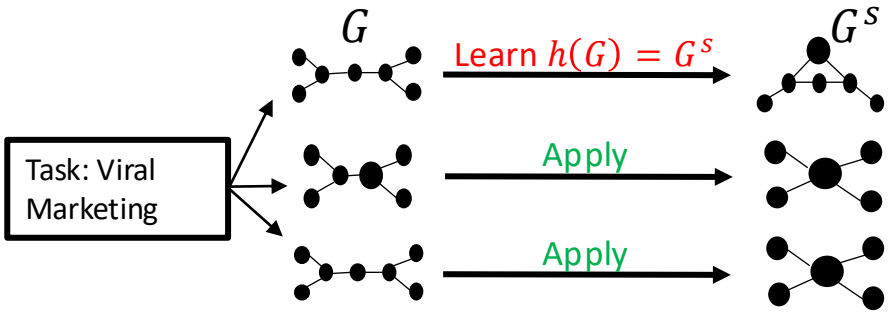
Community Detection	Influence-based Tasks	Sense Making	Graph Query	Immunization	Graph Cut
Metis [Karypis+,1998]	CoarseNet [Purohit+, 2014]	VOG [Koutra+,2014]	GraSS [LeFevre+,2010]		
Spectral Clustering [Luxburg, 2007]	SPINE [Mathioudakis+, 2011]	OntoViz [Shen+,2006]	VNM [Buehrer+, 2008]		

What should we do?

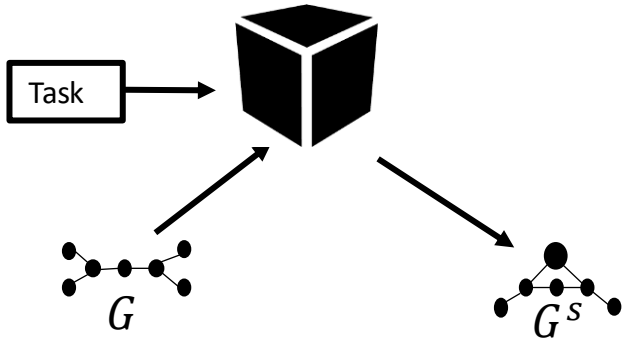
Our Idea: Learn **how** to generate summaries

Why Learning Makes Sense

- Tasks require repeated summary generation



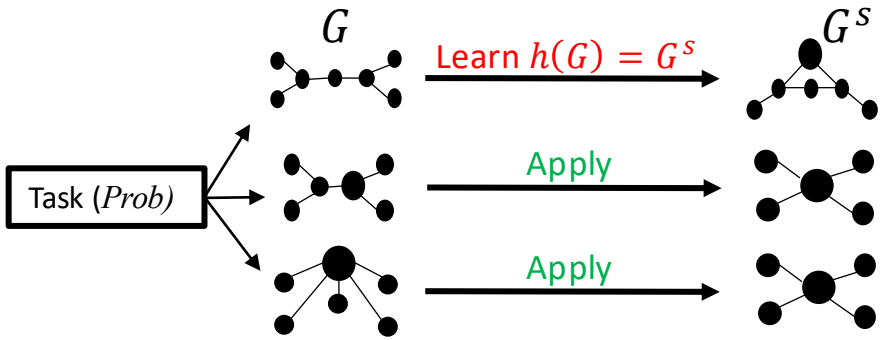
- Objectives for the tasks can be complicated
  - Hard to formulate
  - Domain dependent
  - Black Box



Problem: Learning to Summarize

- **Given** a problem $Prob$ and a distribution D of network instances
- **Learn** how to generate meaningful network summaries that generalizes to unseen instances from D?

Example:
- Similar social networks
- Protein interaction network of similar proteins
- Viral Marketing
- Transportation



More details in the paper

What is the Set of Prob.

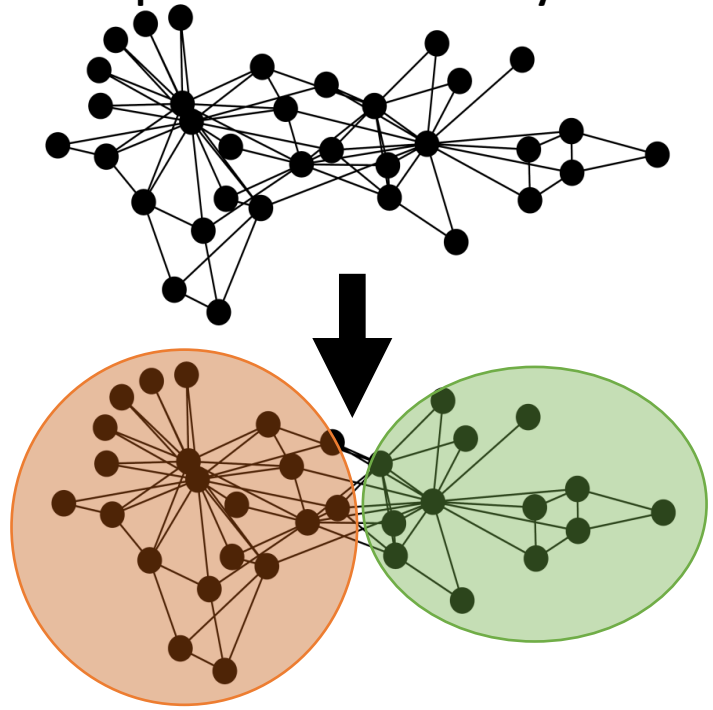
Given a problem ***Prob*** and a distribution D of network instances
Learn how to generate meaningful network summaries that generalizes to unseen instances from D?

- We define a class of problems GOP

Given a graph $G(V, E)$, a set of input parameters Θ and a quality function $F_{prob}(O; G; \Theta) \in R$ and set U
Find the best set of objects O^* such that,

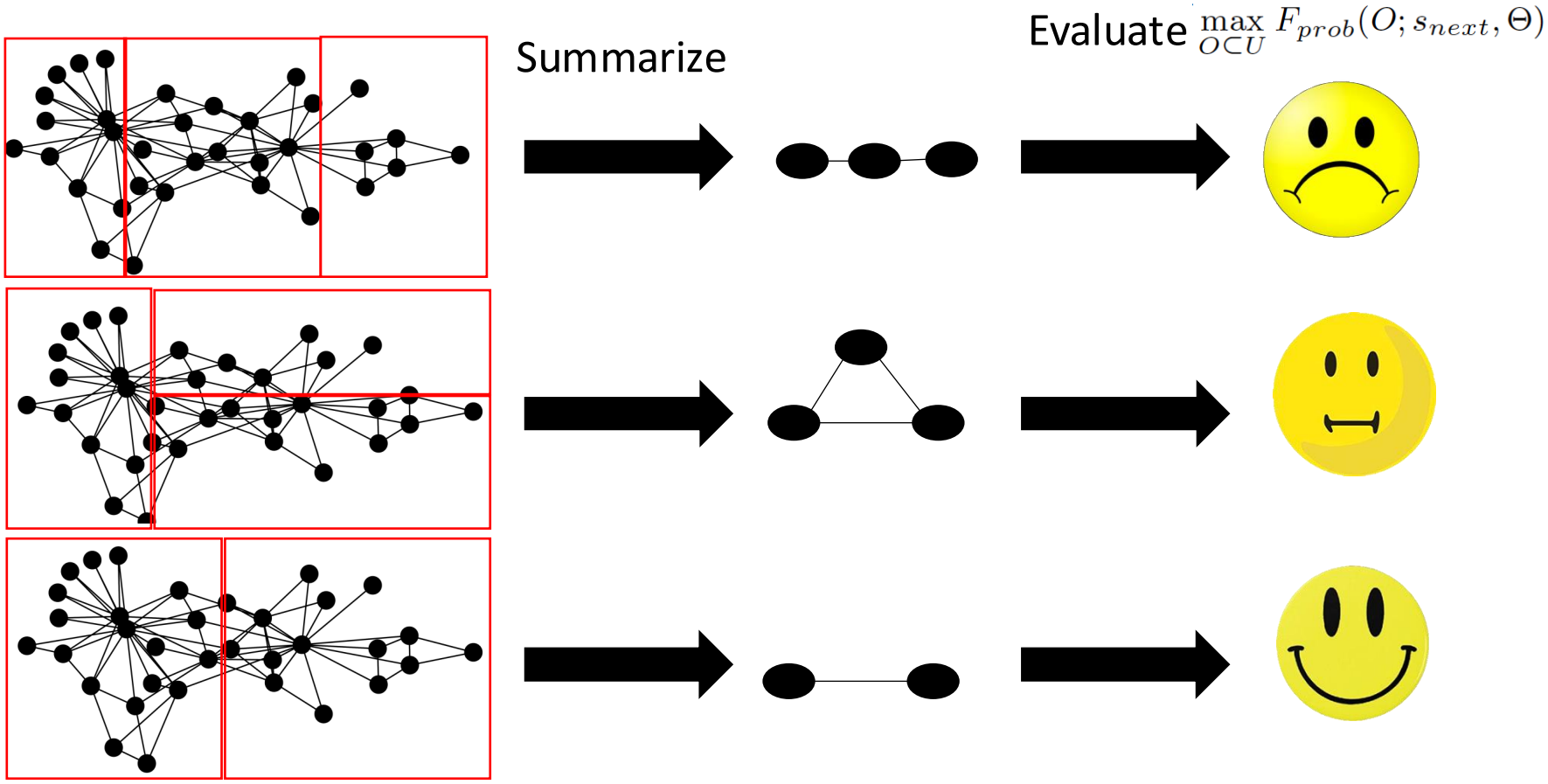
$$O^* = \arg\max_{O \subseteq U} F_{prob}(O; G, \Theta)$$

An example of GOP: Community Detection



Many Tasks are part of GOP
- Immunization
- Ncut
- …

Method: NetGist Overview



NetGist learns *how* to summarize from examples

More details in the paper

NetGist: Reinforcement Learning

Atari Game

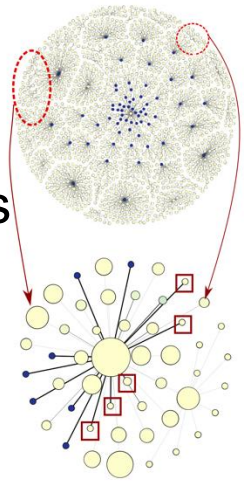
- Universe of Actions
  - left / right
- Universe of States
  - current screen



- Reward
  - score you earned at current step
- Policy
  - How to choose your action
- Learning Procedure
  - Maximize future reward

Network Summarization

- Universe of Actions
  - Merge Nodes
- Universe of States
  - All possible summaries



- Reward
  - Solution to the Task
- Policy
  - How to generate summary
- Learning Procedure
  - Maximize future reward

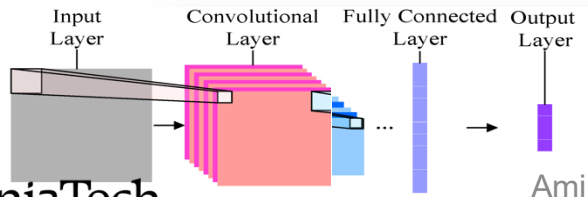
Complete NetGist Algorithm

Algorithm 2 Overview of NetGist

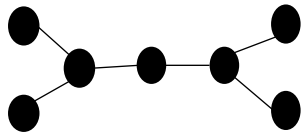
Require: $prob \in$ GOP, α, D

1: Randomly Initialize the deep Q-learning parameters
2: **for** $i = 1$ **To** num of samples **do**
3: Sample $G(V, E)$ and Θ drawn from D
4: // learning how to summarize
5: **for** episode=1 **to** T **do**
6: $G'(V', E') \leftarrow G(V, E)$
7: **while** $|V'| > \alpha \cdot |V|$ **do**
8: Merge $G'(V', E')$ // **Action**
9: // Evaluate **Evaluate**
10: $O', F_{prob} \leftarrow$ DivideAndConquer$(G, G', prob)$ Alg. 1
11: // Optimize
12: Update the deep Q-learning parameters to yield better summary **Update**
13: **Return** the trained model (which solves the TBNS problem)

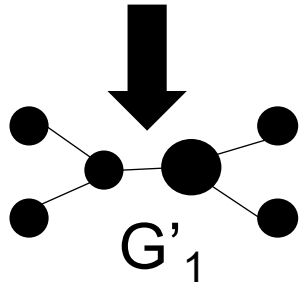
Episode



Summarize



Original Graph



Evaluate

$$\max_{O \subset U} F_{prob}(O; s_{next}, \Theta)$$

Outline

- ☑ Background and Motivation
- ☑ Problem Formulation
- ☑ Method
- ❑ **Experiments**
- ❑ Conclusion

Experiments: datasets

- Various datasets in different scales and domains.

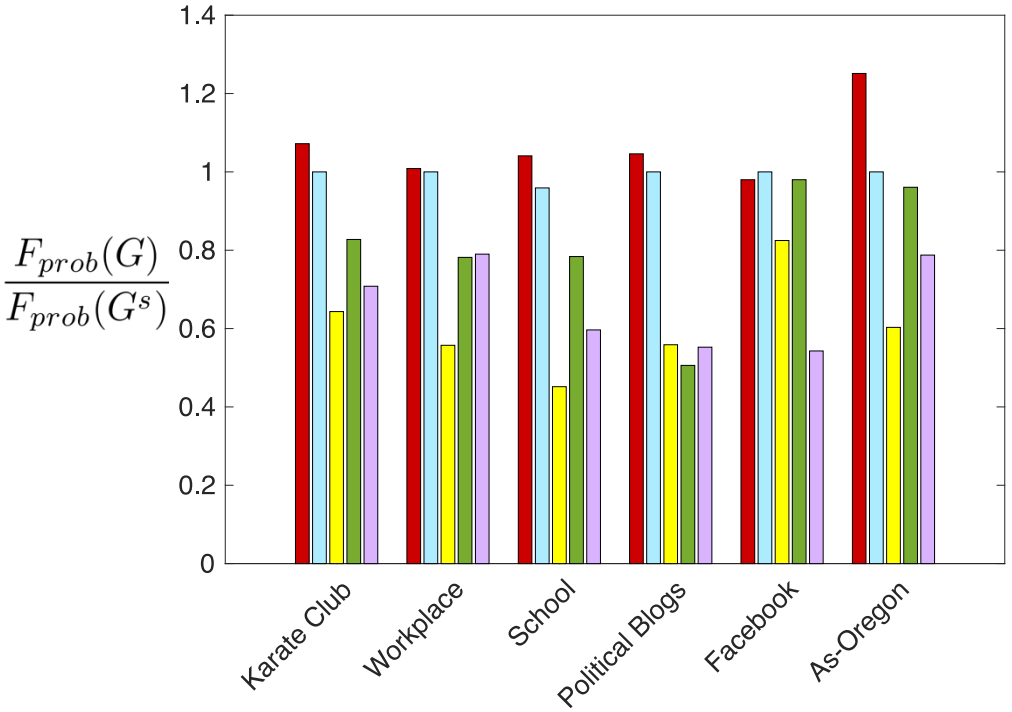
		Dataset	#Nodes	#Edges	
Synthetic	1	ER	20-2,000	50-10,000	
	2	BA	20-2,000	40-4,000	
	3	Block Model	20-2,000	40-30,000	
Real-World	4	Karate Club	34	78	← social network
	5	Work-place	92	755	← mobility networks
	6	High-School	327	5,818	
	7	Political blogs	1,490	16,783	← blogs network
	8	Facebook	4,039	88,234	← social network
	9	As-Oregon	7,000	1,5743	autonomous systems

Experiments: baselines

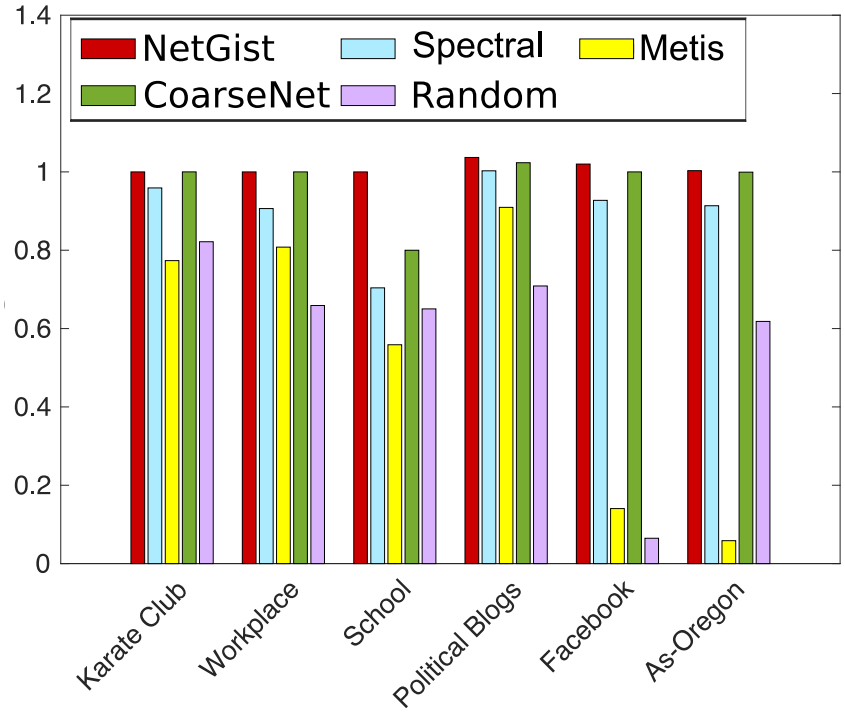
- **Random:** randomly merges nodes
- Other task-specific methods

Community Detection	Influence-based Tasks
Metis [Karypis+,1998]	**CoarseNet** [Purohit+, 2014]
Spectral Clustering [Luxburg, 2007]	

NetGist on Real Networks



Community detection



Influence maximization

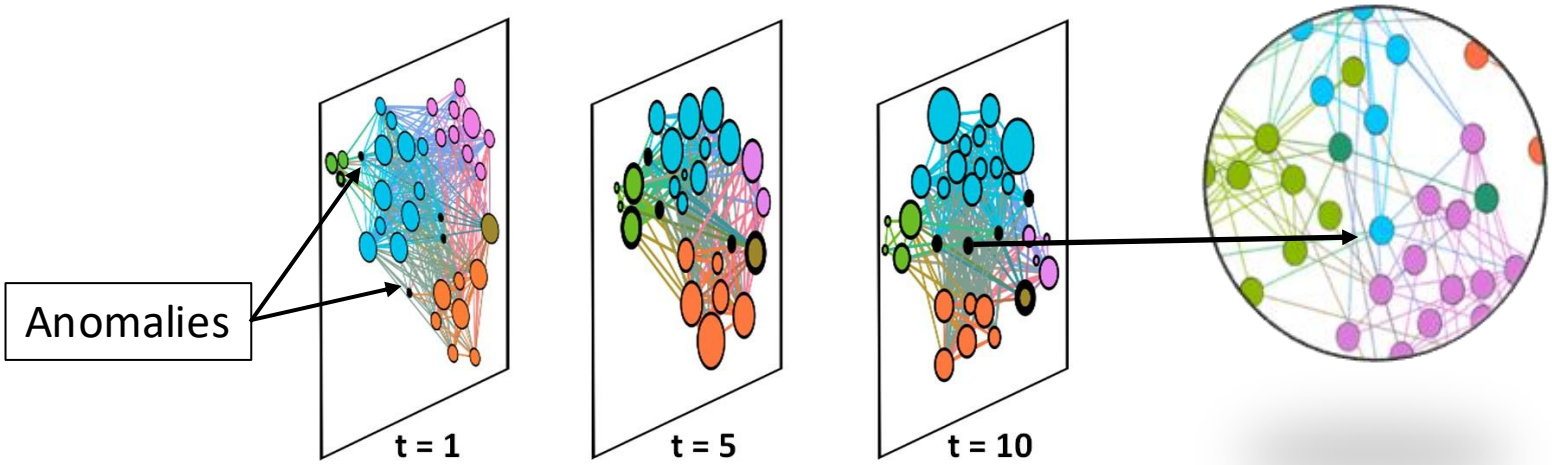
NetGist outperforms the baselines

Additional Results on
1. Varying Distribution
2. Generalization
3. …

In paper

Detecting anomalies on mobility graphs

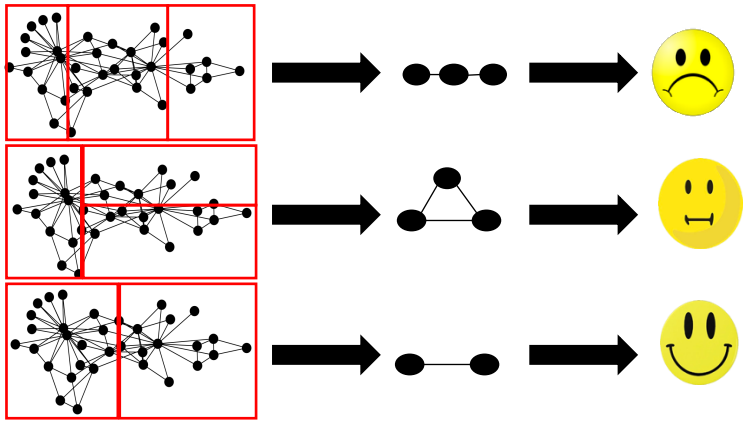
- Work-Place network [Génois+, 2015] , contact network between employees in a work place
- Color represents different departments

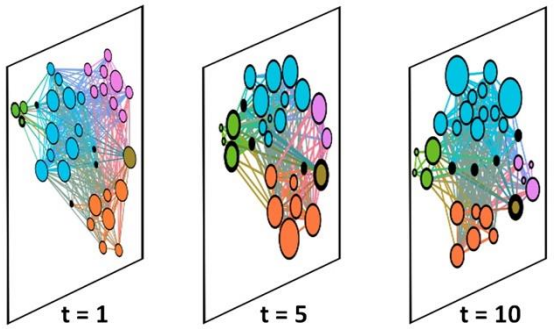


NetGist can detect correct anomalies that match the ground-truth

Conclusion

- Novel Problem and Approach
  - **Learning** to generate **task based summaries**
  - Deep Reinforcement Learning
  - Learns effective summaries
- Applications
  - Anomaly detection
    - Sense making





Thank you!

Code: github.com/SorourAmiri/NetGist






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NSF

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Learning to generate network summaries

